Final Project Guidelines: Introduction to Data Science

Objective

Apply the full data science lifecycle using R Markdown, including data collection, preprocessing, modeling, evaluation, and interpretation.

Project Overview

- Use ONE real-world dataset (minimum 500 rows)
- Choose ONE task: Classification, Regression, or Clustering
- Implement using ONE R Markdown (.Rmd) file
- Ensure code is reproducible and automated

Dataset Requirements

- Must be real-world (Kaggle, UCI, data.gov, etc.)
- Must include both categorical and numerical features
- Provide dataset source and link
- Avoid toy/synthetic datasets

A. Data Collection

- Import dataset automatically (no local paths)
- Ensure code runs on any machine without modification

B. Data Exploration (EDA)

- Show dataset shape, types, and summary statistics
- Identify feature types
- Visualize distributions and correlations
- Detect missing values and outliers

C. Data Preprocessing

- Handle missing values and outliers
- Encode categorical variables
- Normalize/standardize numerical features
- Perform feature engineering
- Reduce skewness if necessary

D. Modeling

- Implement at least one suitable model
- Optional: apply model tuning for bonus

E. Evaluation & Interpretation

- Classification: Accuracy, Precision, Recall, F1-score
- Regression: RMSE, MAE, R^2
- Clustering: Silhouette Score, Visualization
- Provide interpretation via comments and plots

Submission Files

- .Rmd file and corresponding HTML output
- Code must run fully and produce outputs

Evaluation Criteria (50 Marks)

- Data Collection & Description – 5
- EDA & Preprocessing – 15
- Modeling & Evaluation – 15
- Insights & Interpretation – 10
- Code Quality – 5

Deadline & Viva

- Deadline: 2nd May 2026, 11:59 PM
- Viva will be conducted individually

Important Notes

- Late submission: 10% deduction per day
- Use set.seed() for reproducibility
- Creativity and visualization quality may earn bonus marks

Final Project Evaluation Rubric (Total: 50 Marks)

1. Data Collection & Description (5 Marks)

Criteria	Excellent (4–5)	Good (2–3)	Poor (0–1)
Dataset Quality	Real-world, ≥500 rows, appropriate for task	Minor issues (size/type mismatch)	Inappropriate or very small dataset
Source & Description	Clear link, source, and explanation	Partial information	Missing or unclear
Data Loading	Fully automated (URL, reproducible)	Minor issues	Hard-coded / not reproducible

2. Data Exploration & Preprocessing (15 Marks)

Criteria	Excellent (12–15)	Good (7–11)	Poor (0–6)
EDA Completeness	Thorough (structure, summary, correlations, distributions)	Covers basics but lacks depth	Minimal or missing EDA
Visualization	Multiple meaningful plots with interpretation	Some plots, limited explanation	Few or no plots
Data Cleaning	Proper handling of missing values, outliers	Partial handling	Poor or no cleaning
Feature Engineering	Meaningful feature creation/transformation	Basic transformations	No feature engineering
Automation	Generalized preprocessing (not hardcoded)	Some hardcoding	Fully manual

3. Modeling & Evaluation (15 Marks)

Criteria	Excellent (12–15)	Good (7–11)	Poor (0–6)
Model Selection	Appropriate and justified	Reasonable but not justified	Incorrect choice
Implementation	Correct and clean	Minor issues	Incorrect/incomplete
Evaluation Metrics	Correct metrics + clear explanation	Metrics used but weak explanation	Wrong/missing metrics
Validation	Proper train-test split or validation	Basic split only	No validation

4. Insights & Interpretation (10 Marks)

Criteria	Excellent (8–10)	Good (5–7)	Poor (0–4)
Insight Quality	Clear, logical, data-driven conclusions	Some insights but shallow	No meaningful insights
Interpretation	Explains model results clearly	Basic explanation	No interpretation
Visualization Support	Insights supported by visuals	Limited support	No connection to visuals

5. Code Quality & Reproducibility (5 Marks)

Criteria	Excellent (4–5)	Good (2–3)	Poor (0–1)
Readability	Clean, well-structured, commented	Some structure	Messy/unreadable
Reproducibility	Runs fully without modification	Minor issues	Does not run
Automation	Dataset-independent logic	Partial hardcoding	Fully hardcoded